

DR. JUAN DIEGO JARAMILLO SALAZAR

Curriculum Vitae



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Colombia

<i>Email</i>	judijasa@gmail.com
<i>ResearcherID</i>	I-5946-2017
<i>GitHub</i>	judijasa
<i>Place of Birth</i>	Colombia
<i>Date of Birth</i>	28 March 1983
<i>Nationality</i>	Colombian

"The good life is one inspired by love and guided by knowledge."

BERTRAND RUSSELL

POSITIONS

<i>Data Engineer</i> <i>(remote)</i>	2023-2026 mixrank.com Headquarters: 4955 Santa Rita Rd El Sobrante CA 94803, USA Founder/CEO: Scott Milliken	MixRank
<i>Lecturer</i>	2019-2020 Universidad del Valle Sede Tuluá Villa Campestre Calle 43 No. 43-33 Tuluá, Colombia	Universidad del Valle
<i>Postdoc</i>	2018-2019 Institut für Physik Universitätsstr. 1 D-86159 Augsburg, Deutschland Advisor: Prof. Dr. Dr. h.c. Peter Hänggi	Universität Augsburg

Postdoc 2016-2017 National University of Singapore
 Department of Physics
 2 Science Drive 3
 S-117551 Singapore, Singapore
Advisor: Dr. Jiangbin Gong

Postdoc 2015 University of Massachusetts Boston
 Department of Physics
 100 Morrissey Blvd.
 Boston MA 02125-3393, USA
Advisor: Dr. Adolfo del Campo

EDUCATION

PhD 2010-2014 Leibniz Universität Hannover
 Institute for Theoretical Physics
 Appelstrasse 2, 30167 Hannover, Germany
Thesis: *One Dimensional 4-Component Alkali Fermions*
Advisor: Dr. Jun.-Prof. Temo Vekua

Diploma 2008-2009 Abdus Salam Intl. Centre for Theo. Physics
 Condensed Matter Physics Section
 Str. Costiera, 11, Trieste, Italy
Thesis: *Topological Order in Matter*
Advisor: Dr. Prof. Giuseppe Mussardo

Bachelor of Science 2001-2007 Universidad del Valle - Colombia
 Department of Physics
 Calle 13 # 100-00, Cali, Valle del Cauca, Colombia
Thesis: *Local and Global Quantum Computation: Evaluation of Interaction Cost in Nano-structures*
Advisor: Dr. John H. Reina

PUBLICATIONS

A list of my scientific publications can be found at

<https://judijasa.github.io/Publications.html>

COMPUTER SKILLS

Intermediate HTML, CSS, PHP, SQL, BASH (Unix Shell)

Advanced PYTHON, WOLFRAM MATHEMATICA, L^AT_EX

WEB AND SCIENCE PROJECTS

The following are projects involving programming skills. Other projects in which I have participated can be found in my list of scientific publications.

Monitoring Government Websites

I scrap public data (job announcements) from the website of the government of Colombia. The data is filtered and republished in my personal website: *SIMOExpress*. The motivation is to learn scrapping and improve the visualization of government job announcements. The project combines server side and client side programming (PHP, CSS, HTML). To keep the website *SIMOExpress* updated, the program was deployed in a Virtual Machine from Azure Cloud Service. Using CRONJOB, I scheduled the execution of a BASH script where all the (sub)jobs are orderly executed: scrapping data via CASPERJS, storing and filtering data with mySQL and automatic email report via PHPMAILER. Currently, the website *SIMOExpress* is not publicly available due to budget limitations (Azure free service is limited). A working version remains in my personal computer and the code is available in my Github account:

<https://github.com/judijasa/SIMOExpress>

SIMO Express
Ofertas de empleo público en Colombia

Visite la página oficial:
[Sistema de apoyo para la Igualdad, el Mérito y la Oportunidad \(SIMO\)](#)

Buscar por departamento:
-- todos los deptos --

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Palabras clave	Municipio	Salario	Cierre de inscripciones	OPEC
Profesional, Telemática, Ingeniería Industrial, Ingeniería de Sistemas, Ciencias Administrativas, Ciencias Sociales, Ciencias Humanas, Ciencias Políticas, Economía y Finanzas, Leyes y Jurisprudencia, Relaciones Internacionales, Comercio Internacional, Administración Pública, Administración de Empresas, Administración Financiera	• • Cundinamarca	\$276869	2019-10-31	2635
Profesional, Forestal, Pecuaria, Sanitaria, Ingeniería Agronómica, Ingeniería Agrícola, Ingeniería Ambiental	• • Cundinamarca	\$276869	2019-10-31	2636

Virtual Assistant

The government of Colombia has specific protocols to restore the constitutional rights of children. Efficacy in the activation of these protocols is specially important for teachers and directives in educational institutions. This project creates a virtual assistant to quickly register and report situations that affect school coexistence. The virtual assistant is comprised of 5 modules: i) sexual abuse, ii) bullying, iii) consumption of illegal drugs, iv) carrying of weapons and v) selling of illegal drugs. The program is originally designed for mobile devices with internet connection. It has a session for authorised users, allowing them to register and edit situations that affect school coexistence. The program remains incomplete. During the development, the government deployed a program with similar functionalities (SIUCE: Sistema de Información Unificado de Convivencia Escolar). I managed to complete the module on *sexual abuse*, where a form guides the user in order to register the situation. The program uses HTML/CSS and PHP/SQL. It can be publicly accessed at the address:

http://asistente.rf.gd/asistente_rutas_mobile/index.php

The source code is available in my Github account:

<https://github.com/judijasa/mySIUCE>

Quantum Simulation with Python

Quantum entanglement is a nonclassical resource for quantum computation. Bell states are maximally entangled two-qubit states. In L. Magazzù et al 2018 Phys. Scr. 93 064001 we propose a protocol to generate Bell states between two qubits (*B* and *C*) that interact indirectly, through a third, ancillary qubit (*A*). The dynamics is controlled by scheduling repeated projective measurements on the ancillary qubit every time τ . With the purpose to quantify the protocol's

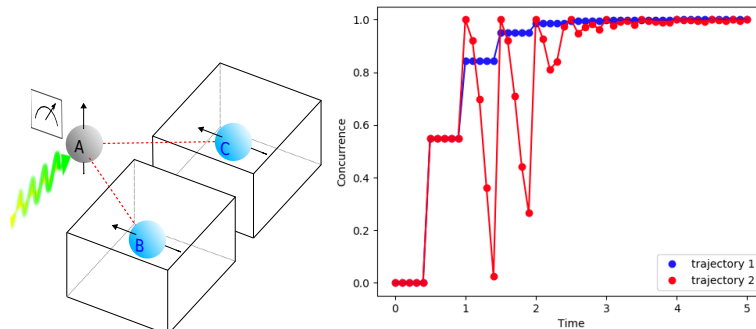
ability to generate Bell States I compute the Concurrence (a way of measuring quantum entanglement) of the mixed state resulting from the partial trace of the ancillary qubit (A). This is done for every instant of time during the protocol. The mixed states of the overall system are provided by the first author (L. Magazzù) in a .dat file. My code is written in **Python**. I import the data using the function `genfromtxt()` from the **Numpy** library. The figures are generated with **Matplotlib**. To represent and process the quantum states, I use the **QuTip** library:

J. R. Johansson, P. D. Nation, and F. Nori: “QuTiP 2: A Python framework for the dynamics of open quantum systems.”, *Comp. Phys. Comm.* 184, 1234 (2013) [DOI: [10.1016/j.cpc.2012.11.019](https://doi.org/10.1016/j.cpc.2012.11.019)].

J. R. Johansson, P. D. Nation, and F. Nori: “QuTiP: An open-source Python framework for the dynamics of open quantum systems.”, *Comp. Phys. Comm.* 183, 1760-1772 (2012) [DOI: [10.1016/j.cpc.2012.02.021](https://doi.org/10.1016/j.cpc.2012.02.021)].

The following are snapshots of the code and visualization:

```
rho44=[]
rho14=[]
min=1 # min/max rho's
max=51
range_input=2**3
# total 201
for k in range(min-1, max):
    gx.append(k/10)
    a=np.genfromtxt('new_data2/Re_rho_t-ntrj=1-tau=0.5-Jx=1-Jy=0-Jz=0-g1=0-gphi=0-i')
    b=np.genfromtxt('new_data2/Im_rho_t-ntrj=1-tau=0.5-Jx=1-Jy=0-Jz=0-g1=0-gphi=0-i')
    f = []
    for i in range(range_input):
        for j in range(range_input):
            bvar1=debi(i)
            bvar2=debi(j)
            a1, b1, c1 = bvar1
            a2, b2, c2 = bvar2
            ket_i = tensor(basis(2,int(a1)),basis(2,int(b1)),basis(2,int(c1)))
            bra_j = tensor(basis(2,int(a2)),basis(2,int(b2)),basis(2,int(c2))).dag()
            f.append(complex(a[i][j],b[i][j])*ket_i*bra_j)
    rho=sum(f)
    subrho=rho.ptrace([1,2])
    gy.append(concurrence(subrho))
```



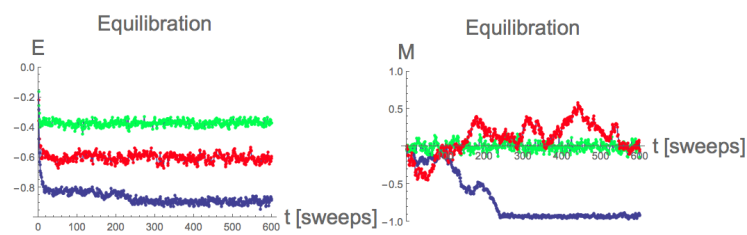
Ising Model and the Travelling Salesman

This project supports the Computational Physics component of the lecture in Statistical Mechanics given by Prof. Dr. Holger Frahm at the Leibniz University of Hannover. In a Wolfram Mathematica Notebook I implement the **Metropolis algorithm** to simulate the Ising model, describing the emergent properties of (anti)ferromagnets. The expectation value of observables is estimated using the **Monte Carlo technique**. In the second part of the project I implement **Steepest Descent** and later **Simulated Annealing** to provide an approximated solution to the Travelling Salesman problem. This exploits the analogy between the Ising model and the Travelling Salesman, where the energy is associated to total path length (“tour distance”) and the spin flips (“sweeps”) to path permutations. The Notebooks are written in an expository way, as these are all well known problems and techniques. The following are snapshots of the code and visualization:

```
(* Local (interaction) energy under closed boundary conditions *)

IntBulk[i_, j_] := s[[i, j]] (s[[i, j + 1]] + s[[i, j - 1]] + s[[i + 1, j]] + s[[i - 1, j]])

IntBound[i_, j_] :=
  s[[i, j]] (s[[i, Mod[j + 1, Lsize, 1]]] + s[[i, Mod[j - 1, Lsize, 1]]] +
    s[[Mod[i + 1, Lsize, 1], j]] + s[[Mod[i - 1, Lsize, 1], j]])
```



Figs.1: Equilibration of Energy E[t] and Magnetization M[t] along 600 Metropolis-steps for temperatures T=0.48 (blue), 0.60 (red) and 0.8 (green). Near T=0.3 magnetic fluctuations remain in the scale of the total magnetization.

Number of cities = 20
Computation Time = 0.1118452 min
Minimal tour Distance = 4.38333 (best result found)
Found at 1 out of 5 trials.

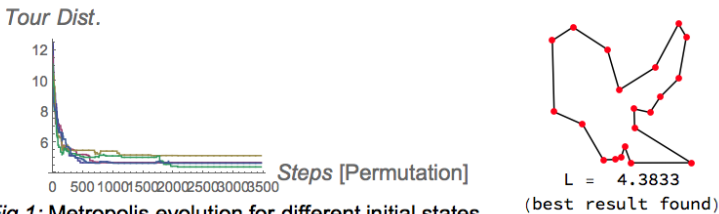


Fig.1: Metropolis evolution for different initial states.

PROFESSIONAL EXPERIENCE

Education	Teaching assistant at Universidad del Valle Teaching assistant at Leibniz Universität Hannover Teaching assistant at Augsburg Universität
Referee	Scientific Reports, Nature Publishing Group New Journal of Physics, IOP Publishing

PROFESSIONAL INTERESTS

General	Science and Technology
Specific	Scientific Software, Web Development

OTHER INFORMATION

Languages	SPANISH · Mothertongue ENGLISH · Fluent GERMAN · Basic
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